

Crafting a New MIAC Basketball Schedule



Alex McKeever

Agenda

- Preliminaries
 - Schedule Requirements
 - Construction Methodology
- Non-Conference Games
- Conference Games
- Evaluation Metrics
 - Bye Weeks
 - Home-Away Splits
 - Geographical Measures / Team Pain
- Some Proposed Schedules

Preliminaries

Schedule Requirements

11 MIAC Teams

Conference: Each team plays each other team

Non-Conference: 6 games (with intent to minimize distance)

If Team A plays @ Team B in the conference games, then Team B plays @ Team A if they meet in the non-conference games

Example Schedule Timeline

We require 3 teams to be on bye for two of the non-conference weeks to ensure each team gets a non-conference bye.

Week	Con	Non-Con	Teams on Bye	Games
Week 1		12/1	1 on bye	5
Week 2	12/4		1 on bye	5
Week 3		12/8	1 on bye	5
Week 4	12/11		1 on bye	5
Week 5		1/5	3 on bye	4
Week 6	1/8		1 on bye	5
Week 7		1/12	1 on bye	5
Week 8	1/15		1 on bye	5
Week 9	1/19		1 on bye	5
Week 10	1/22		1 on bye	5
Week 11		1/26	1 on bye	5
Week 12	1/29		1 on bye	5
Week 13		2/2	3 on bye	4
Week 14	2/5		1 on bye	5
Week 15	2/9		1 on bye	5
Week 16	2/12		1 on bye	5
Week 17		2/16	1 on bye	5
Week 18	2/19		1 on bye	5

Construction Methodology

There are billions of possible schedules...

Generate a bunch of schedules and evaluate them using various metrics (i.e. do they make a “fair” schedule)!

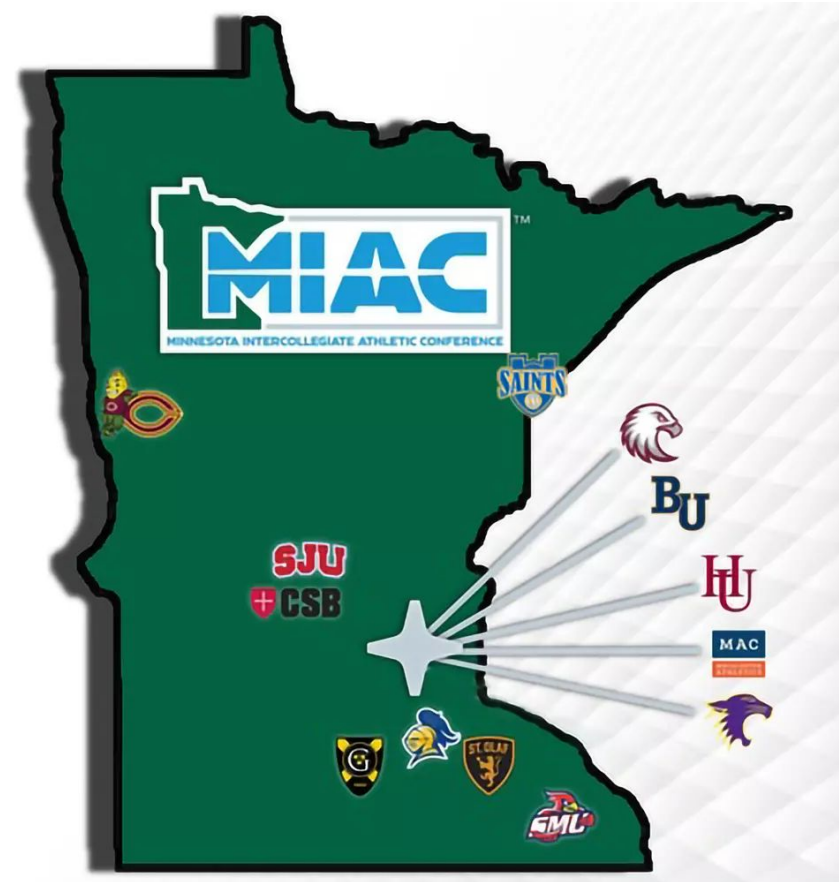
- Non-Conference
 - Randomize who plays who (to some extent)
 - Randomize schedule and home/away
- Conference
 - Randomize schedule and home/away
- Evaluate

Non-Conference

Selecting the Games

Drive Time Matrix		AUG	BET	CAR	CON	GAC	HAM	MAC	SJU	SMU	STO	CSS
Augsburg	AUG		18	51	212	72	13	13	78	127	48	143
Bethel	BET	18		60	209	81	16	20	75	131	57	134
Carleton	CAR	50	59		252	61	53	50	118	103	8	179
Concordia	CON	212	210	250		259	217	218	143	330	247	255
Gustavus	GAC	73	83	60	257		80	77	124	137	59	207
Hamline	HAM	12	16	53	217	79		11	83	125	50	141
Macalester	MAC	13	21	50	219	77	11		85	125	48	143
Saint John's	SJU	77	74	115	142	123	81	83		195	112	167
Saint Mary's	SMU	126	132	101	330	137	124	126	196		105	228
St. Olaf	STO	46	56	7	249	60	49	47	115	105		177
St. Scholastica	CSS	142	134	179	255	205	141	143	168	230	178	

We create a drive time matrix that records the approximate time to travel from each college to every other college. This matrix will help us decide which non-conference games should be played.



Selecting the Games

Fix the games that make the most sense based on drive time

- Carleton and St. Olaf
- City teams play other city teams

Never allow the games that are between far colleges

- Scholastica and Concordia

Randomize other opponents to some extent

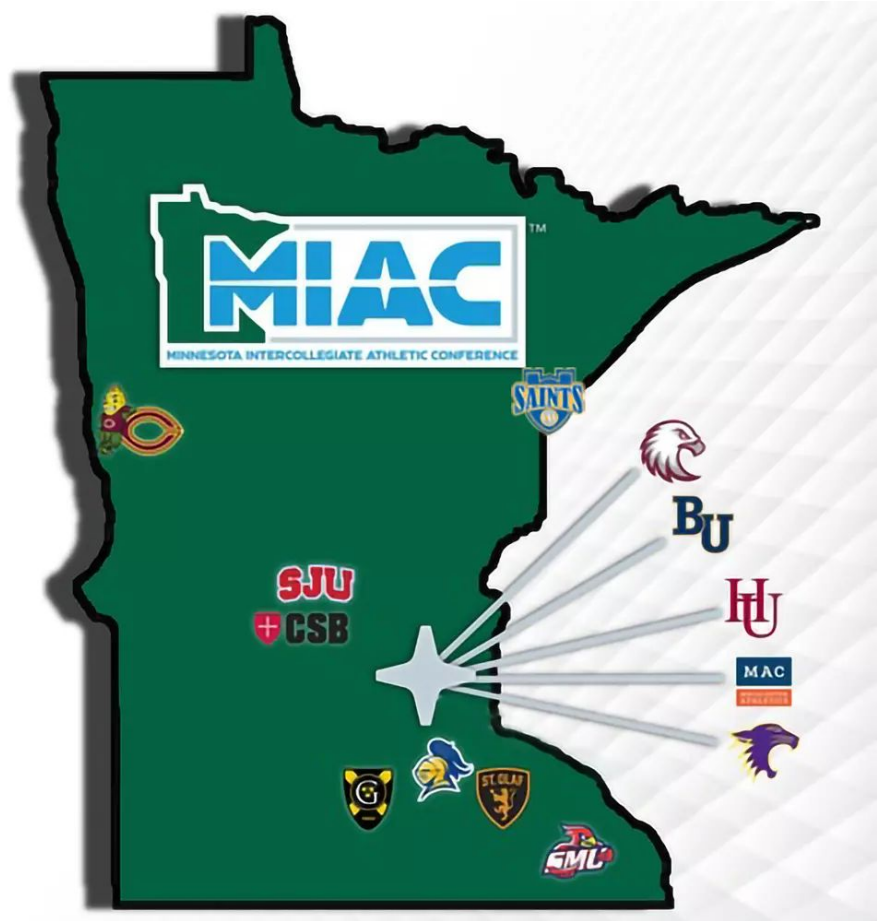
AUG	CSS	2 of			3 of					
		BET	HAM	MAC	SMU	CON	SJU	CAR	STO	GAC
BET	CSS	2 of			3 of					
		AUG	HAM	MAC	SMU	CON	SJU	CAR	STO	GAC
HAM	CSS	2 of			3 of					
		BET	AUG	MAC	SMU	CON	SJU	CAR	STO	GAC
MAC	CSS	2 of			3 of					
		BET	HAM	AUG	SMU	CON	SJU	CAR	STO	GAC
CAR	STO	GAC	SMU	3 of						
				AUG	BET	HAM	MAC	SJU	CON	CSS
STO	CAR	GAC	SMU	3 of						
				AUG	BET	HAM	MAC	SJU	CON	CSS
GAC	STO	CAR	SMU	3 of						
				AUG	BET	HAM	MAC	SJU	CON	
SMU	STO	GAC	CAR	3 of						
				AUG	BET	HAM	MAC			
CON	SJU	2 of			3 of					
		CAR	STO	GAC	AUG	BET	HAM	MAC		
SJU	CON	2-3 of				2-3 of				
		AUG	BET	HAM	MAC	CAR	STO	GAC	CSS	
CSS	AUG	BET	HAM	MAC	2 of					
					CAR	STO	SJU			

Carleton will play STO, GAC, and SMU and three other randomly selected teams.

Home and Away

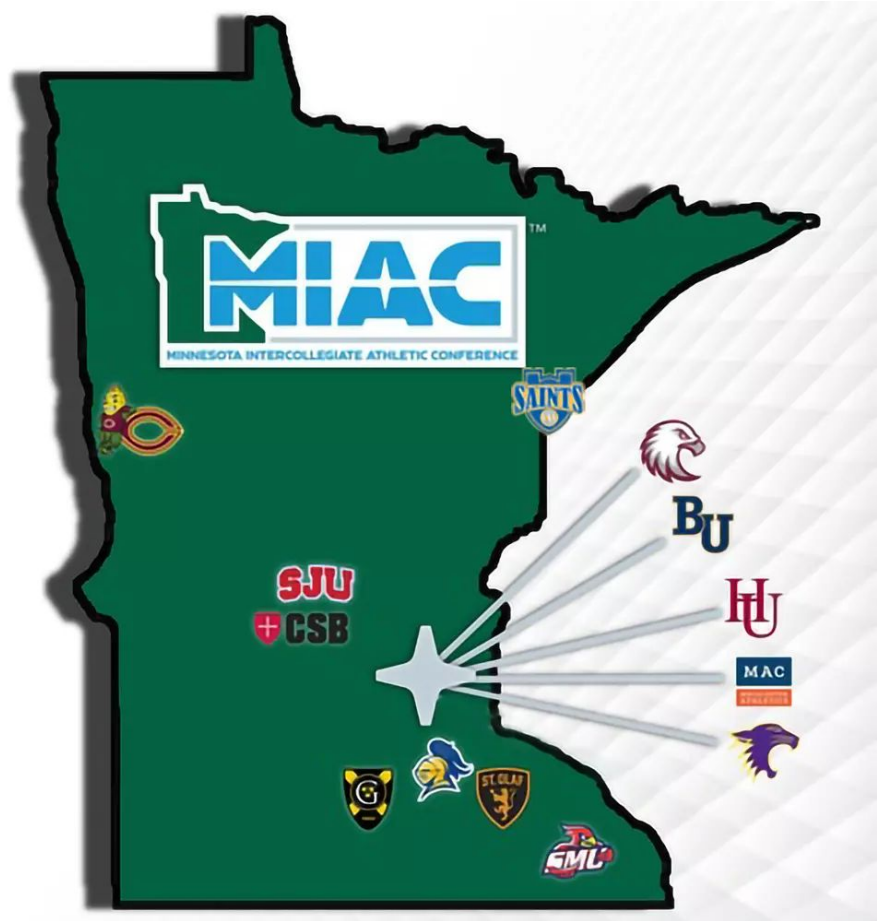
Tough to figure out fairness since some teams will have to travel more miles in a season simply based on geography

However, it is very easy to generate and later inspect once we have hundreds of thousands of schedules to compare



Scheduling the Games

Simply randomize who plays when!

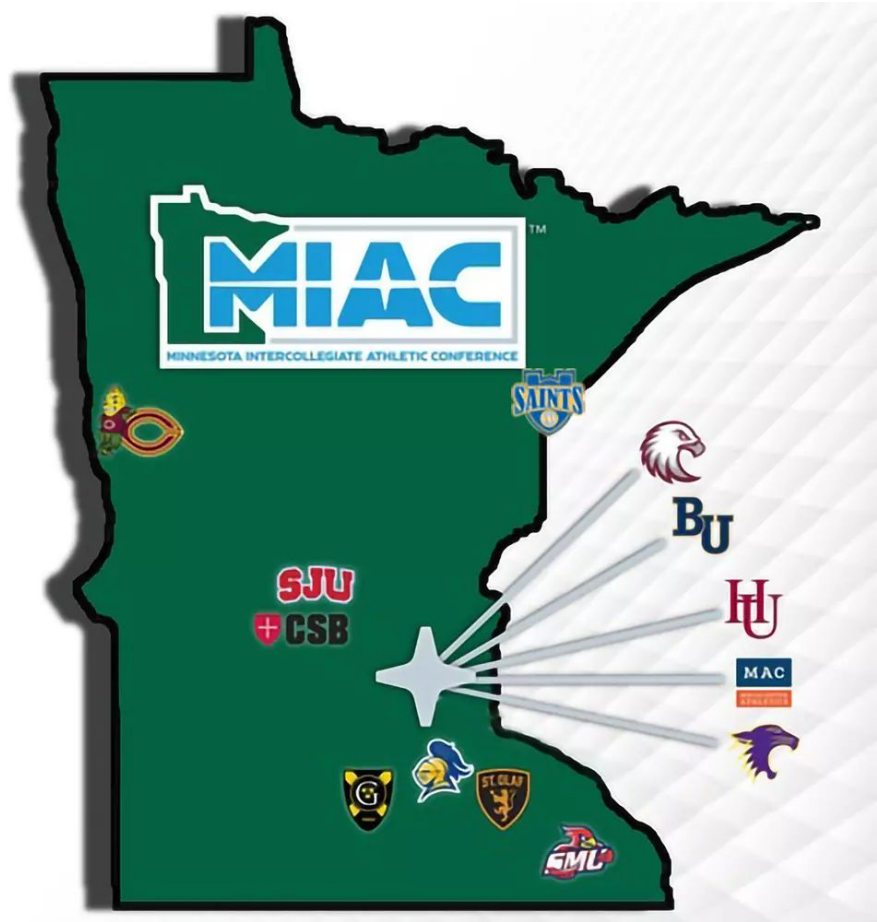


We will examine how good/fair these randomized schedules are in a bit.

Conference

Scheduling the Games

Simply randomize who plays when!



Home and Away

Ensure home/away are flipped from the non-conference games

Randomize the remaining four games for each team

AUG	CSS	2 of			3 of					
		BET	HAM	MAC	SMU	CON	SJU	CAR	STO	GAC
BET	CSS	2 of			3 of					
		AUG	HAM	MAC	SMU	CON	SJU	CAR	STO	GAC
HAM	CSS	2 of			3 of					
		BET	AUG	MAC	SMU	CON	SJU	CAR	STO	GAC
MAC	CSS	2 of			3 of					
		BET	HAM	AUG	SMU	CON	SJU	CAR	STO	GAC
CAR	STO	GAC	SMU	3 of						
				AUG	BET	HAM	MAC	SJU	CON	CSS
STO	CAR	GAC	SMU	3 of						
				AUG	BET	HAM	MAC	SJU	CON	CSS
GAC	STO	CAR	SMU	3 of						
				AUG	BET	HAM	MAC	SJU	CON	
SMU	STO	GAC	CAR	3 of						
				AUG	BET	HAM	MAC			
CON	SJU	2 of			3 of					
		CAR	STO	GAC	AUG	BET	HAM	MAC		
SJU	CON	2-3 of				2-3 of				
		AUG	BET	HAM	MAC	CAR	STO	GAC	CSS	
CSS	AUG	BET	HAM	MAC	2 of					
					CAR	STO	SJU			

Evaluation Metrics on 1.6 Million Schedules

	N1	C1	N2	C2	N3	C3	N4	C4	C5	C6	N5	C7	N6	C8	C9	C10	N7	C11		
																				
																				
																				
																				
																				
																				
																				
																				
																				
																				

[Home](#)
[Away](#)
[Bye](#)

An example generated schedule that is less than ideal. However, we can generate millions of schedules and filter to the ones where these issues do not occur!

Team Pain

$$\text{Team Pain} = (\quad 100 * | \text{Games Between Byes} - 8 | \\ + \quad \text{Conference Distance} * 1.2 \\ + \quad \text{Non-Conference Distance} \\ + \quad 100 * \text{Max Away Streak}) / 100$$

Team Pain is meant to be a metric that captures how difficult a schedule is. This can be expanded to include more measures as well including strength of schedule or if a team plays 4 out of 5 consecutive games as the away team.

Other Metrics

- **RMSE_Total_Dist**
 - Captures the fairness of distance traveled across all games (lower is more fair)
- **RMSE_NonCon_Dist**
 - Captures the fairness of distance traveled across non-conference games (lower is more fair)
- **RMSE_Con_Dist**
 - Captures the fairness of distance traveled across conference games (lower is more fair)
- **RMSE_TeamPain**
 - Captures the overall fairness of the schedule (lower is more fair)
- **AvgPainFromMean**
 - Captures the overall ease of the schedule (lower is easier schedule)
- **AvgNonConDistFromMean**
 - Captures the overall distance traveled (lower is less distance)

The RMSE metrics capture variability. The lower they are, the more equitable the schedule is.

To calculate RMSE we first take the average of the metric across all 11 teams. We then calculate the difference between the team and average and square it. We sum these squared values and then take the square root. The square/square root process helps to flag when island schools have hard schedules.

The average from mean metrics capture overall “goodness”. The lower they are, the more we are minimizing distance (cost) or bad schedule instances like consecutive byes.

To calculate average from mean we first take the average of the metric across all generated schedules for each team. We then calculate the difference between the metric and the average for each team. We then take the averages of these differences across all 11 teams.

Proposed Schedules

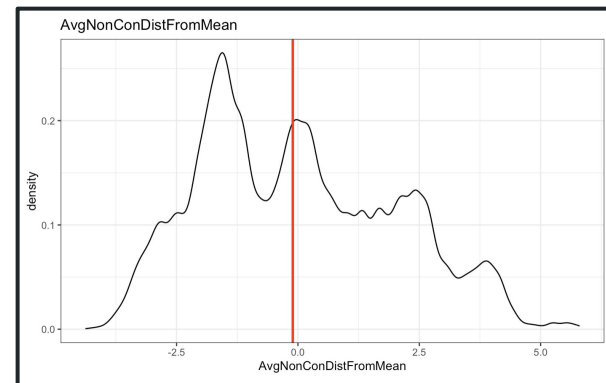
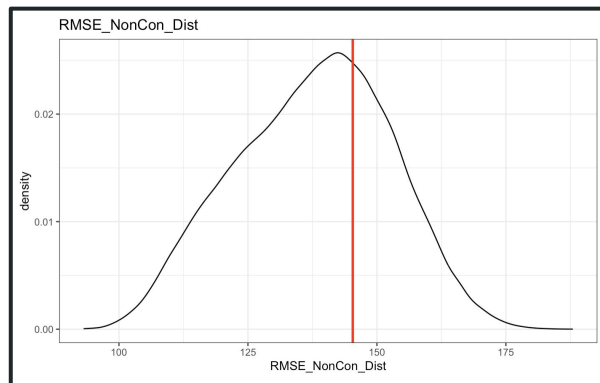
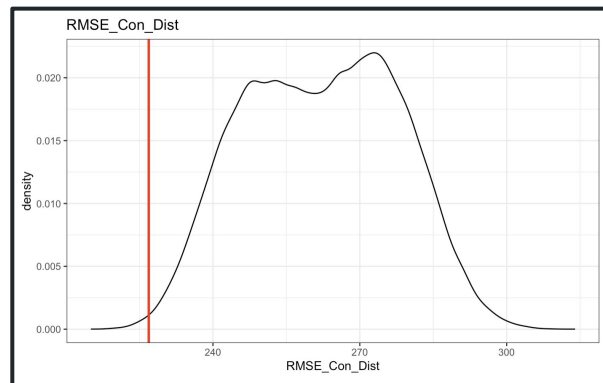
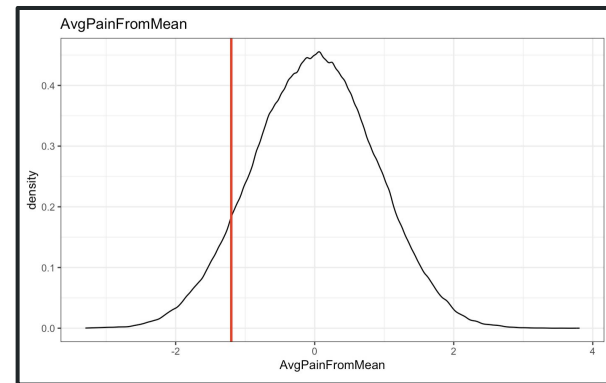
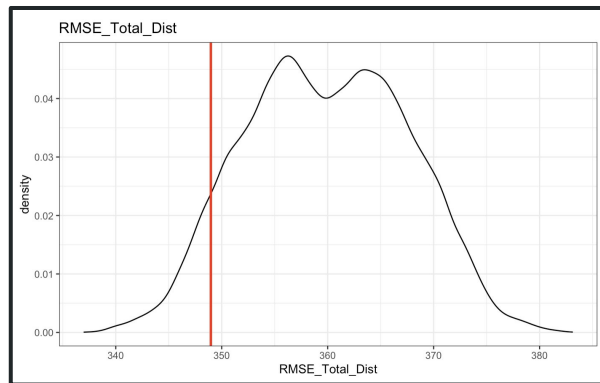
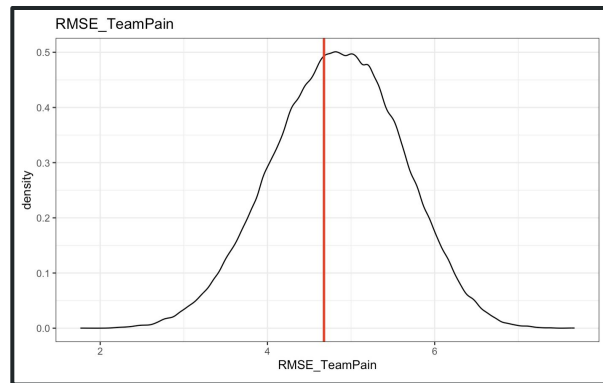
Schedule 1 - 746595

	N1	C1	N2	C2	N3	C3	N4	C4	C5	C6	N5	C7	N6	C8	C9	C10	N7	C11
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		

- Home
- Away
- Bye

A better generated schedule. One can adjust the schedule as well for fine-tuning purposes.

Fairness of Schedule



Being further to the left indicates a more fair or “more good” schedule.

Schedule 2 - 364229

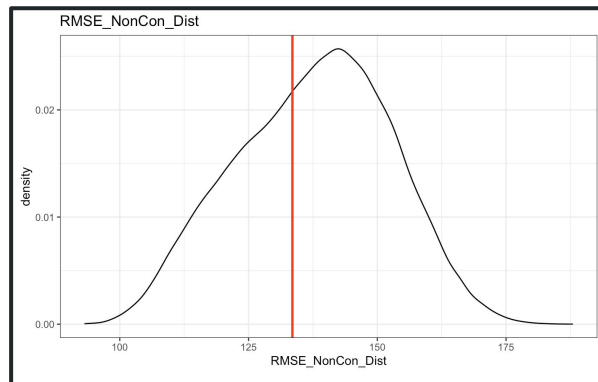
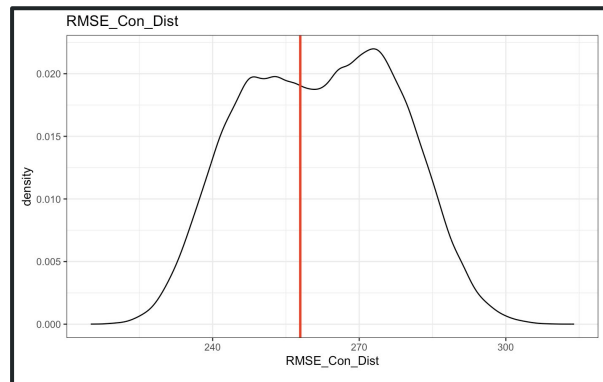
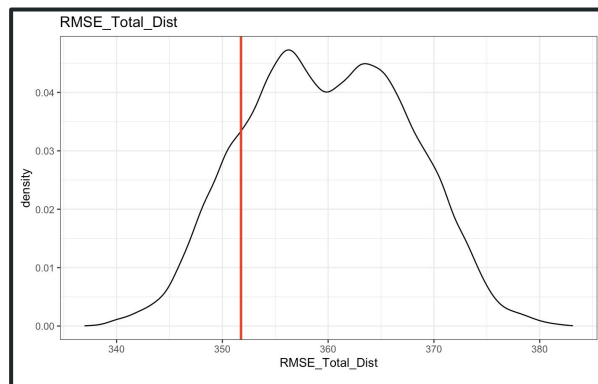
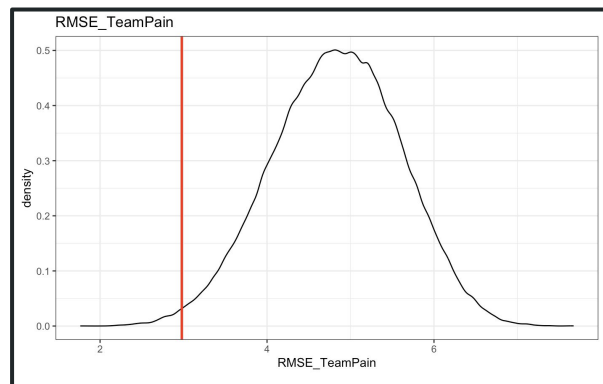
	N1	C1	N2	C2	N3	C3	N4	C4	C5	C6	N5	C7	N6	C8	C9	C10	N7	C11
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		

Home

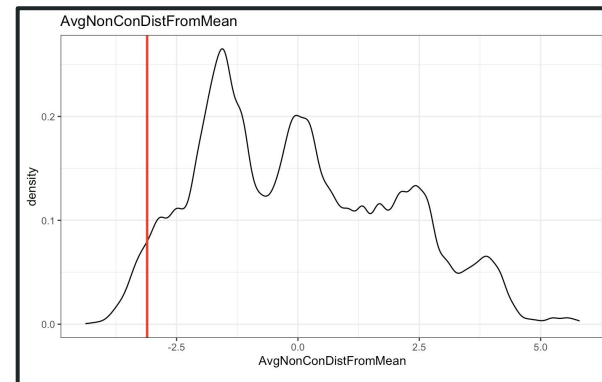
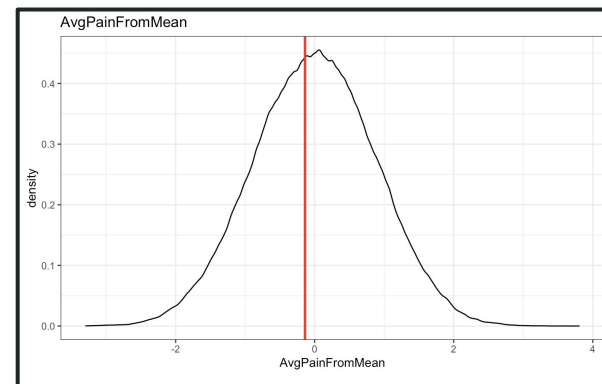
Away

Bye

Fairness of Schedule



“Goodness” of Schedule



Schedule 3 - 17720

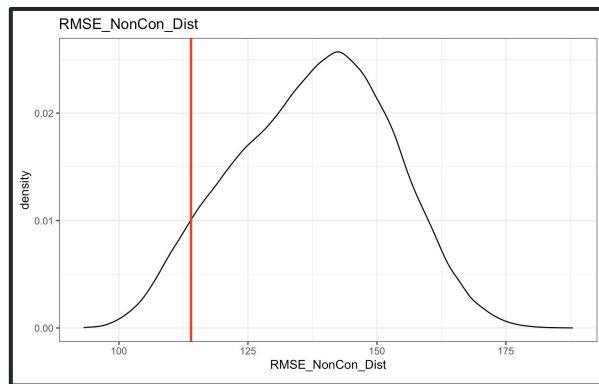
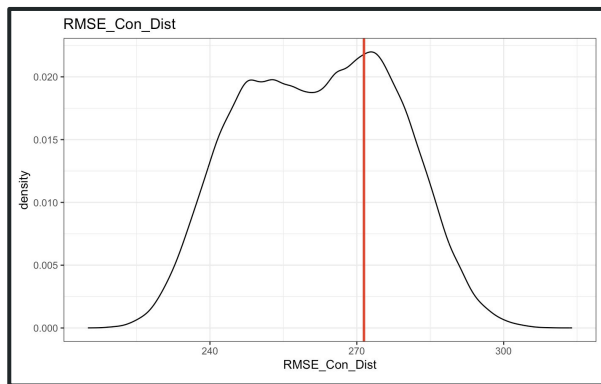
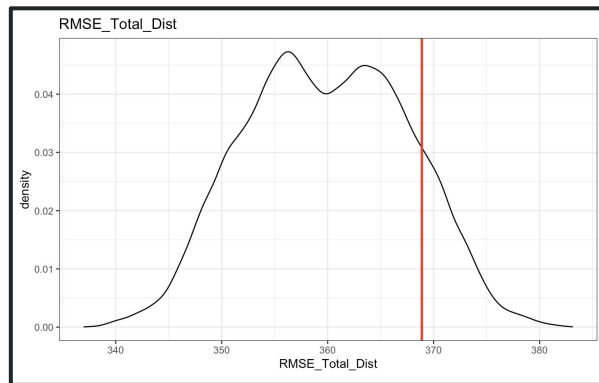
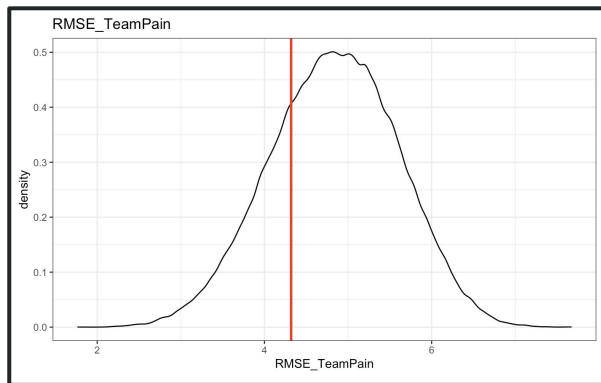
	N1	C1	N2	C2	N3	C3	N4	C4	C5	C6	N5	C7	N6	C8	C9	C10	N7	C11
	BU	SJU		BU														
BU									SJU				SJU					
	SJU							BU				SJU						
							SJU				BU			BU				SJU
						SJU					SJU				BU			
							BU	SJU				BU						
										BU				SJU			SJU	
SJU									BU				BU					
										SJU								
			BU	SJU		BU												
		BU			SJU										SJU		BU	

Home

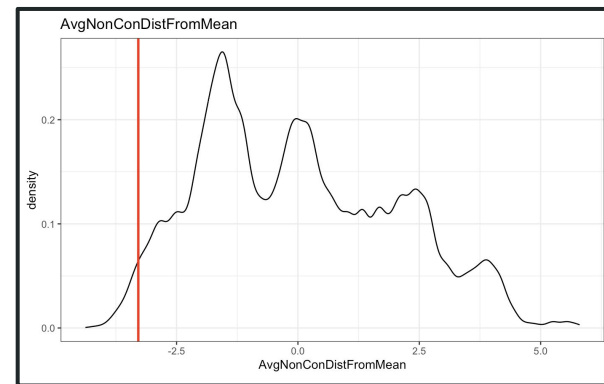
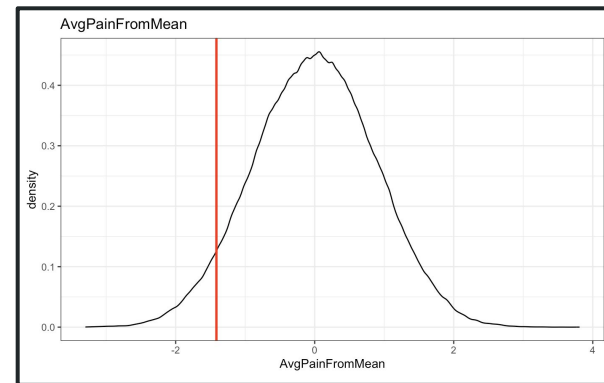
Away

Bye

Fairness of Schedule



“Goodness” of Schedule



Schedule 4 - 109201

	N1	C1	N2	C2	N3	C3	N4	C4	C5	C6	N5	C7	N6	C8	C9	C10	N7	C11
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		

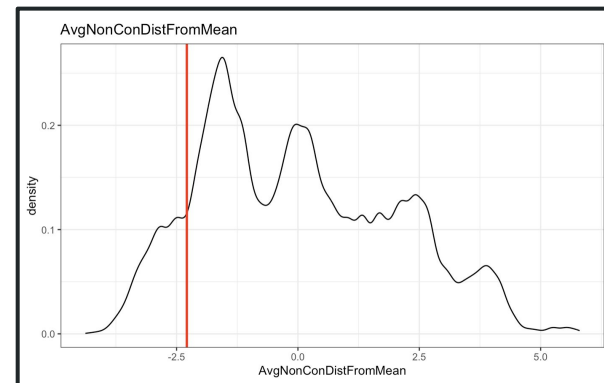
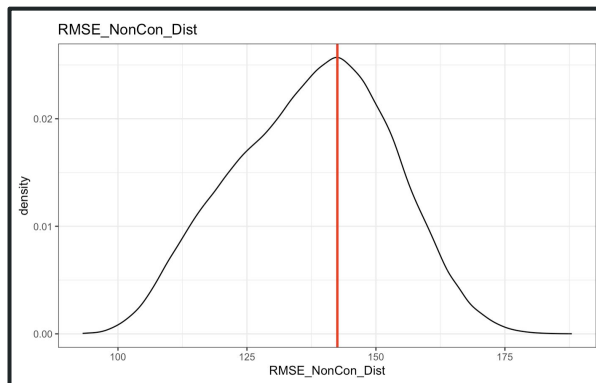
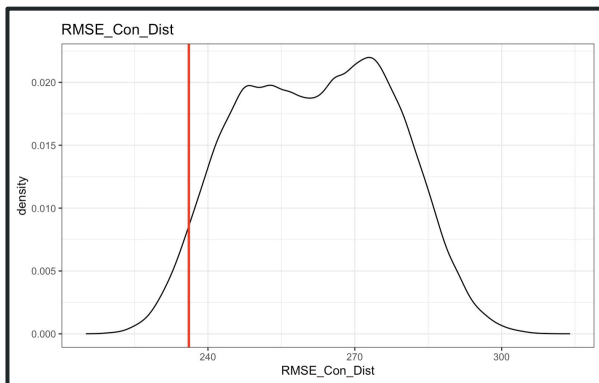
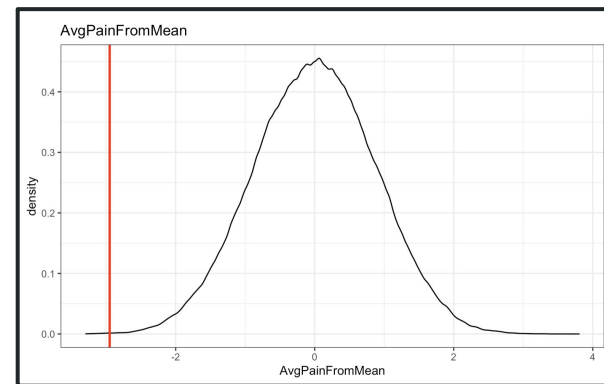
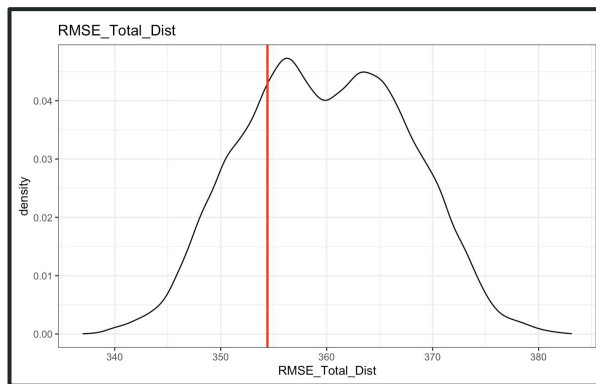
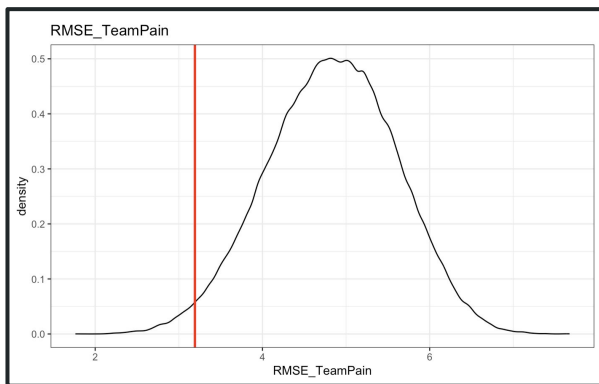
Home

Away

Bye

Fairness of Schedule

“Goodness” of Schedule



Next Steps

- **Implement additional factors into Team Pain**
 - Controlling for instances of a team playing n away games in a m game span
 - Potential addition of some form of strength of schedule
- **Investigate implementation of Women's Basketball**
 - Conference DHs on (mainly) Saturdays
 - Creation of a non-conference schedule with 7 games
- **Potential work on coding extra restrictions to accommodate for things like venue or extracurricular conflicts on certain days**
- **5 week gaps between repeat opponents**
- **Home game in last 2-3 weeks**

	N1	C1	N2	C2	N3	C3	N4	C4	C5	C6	N5	C7	N6	C8	C9	C10	N7	C11
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		
																		

Home

Away

Women's
Schedule